

Unit II

UNDERSTANDING HUMAN MOVEMENT

Introduction

This unit explores the fundamental aspects of human movement, focusing on three key areas: **movement education**, **movement competency**, and **the science of the skeletal and muscular systems**. Movement education emphasizes learning how to move efficiently and effectively. It involves understanding basic movement patterns and how they contribute to overall physical ability.

By developing competency, individuals can enhance their coordination and adaptability in different movement tasks. This includes how bones, muscles, and joints work together to create motion. Understanding these systems helps us comprehend biomechanics and physiological processes involved in human movement. It highlights the practical and theoretical implications of movement education, competency development, and the scientific foundations of our skeletal and muscular systems in Physical Education.

General Objectives:

After studying this Unit, you will be able to:

- Explore how movement education has evolved from its inception in the early 20th century to its current interdisciplinary nature, integrating fields such as kinesiology, psychology, and sociology;
- Investigate key theoretical frameworks in movement education, including Rudolf Laban's Movement Analysis and the Skill Theme Approach, and their applications in promoting physical literacy and skill development across different age groups;

- Foster proficiency in fundamental movement skills essential for physical activities, such as running, jumping, and throwing, and apply movement concepts like biomechanics and motor learning principles to enhance skill acquisition;
- Emphasize movement education as a tool for holistic development, encompassing physical health, emotional well-being, and social interaction through creative and expressive movement experiences; and
- Illustrate how movement competency contributes to lifelong physical activity by discussing its role in enhancing performance, preventing injuries, and maintaining overall well-being across various stages of life.

Introduction to Movement Education

Movement education, introduced in the early 20th century by Rudolf Laban, has undergone significant transformation influenced by a range of factors including artistic expression and scientific inquiry. Originally focused on understanding the qualitative aspects of human movement, this field has evolved into a multidisciplinary approach that incorporates insights from *kinesiology, biomechanics, motor learning, psychology, sociology, and education*. This broad perspective enriches our understanding of how movement contributes to physical health, athletic performance, and injury prevention (Dewey, 2013; Gagen & Getchell, 2014).

Movement education adopts a systematic method for teaching physical movements. Its primary goal is to develop fundamental motor skills, improve coordination, enhance balance, and foster increased awareness of the body. By equipping individuals with the ability to perform movements effectively in various contexts, movement education not only enhances physical abilities but also supports cognitive and emotional aspects related to movement expression.

Through its integrated approach, movement education promotes personal growth and development by catering to diverse needs and abilities across different age groups and environments. It serves as a valuable tool for empowering individuals to achieve

greater proficiency in movement, contributing to overall well-being and quality of life.

Movement Competency

Movement competency is essential for developing proficiency in physical activities by mastering fundamental movement skills and applying movement concepts and strategies. These foundational elements are important for achieving physical literacy, enabling individuals to adeptly perform and adjust movements across various activities.

Fundamental movement skills such as *running, jumping, throwing, and catching* serve as fundamental components for more refined movements. Mastery of these skills not only enhances physical abilities but also establishes a solid foundation for life-long engagement in physical activities. Through consistent practice and guidance, individuals cultivate competence in executing these movements efficiently and with precision.

Movement competency fosters holistic development beyond physical performance. By integrating fundamental skills with conceptual understanding and strategic thinking, individuals acquire the necessary skills to participate in physical activities confidently and effectively. Movement competency, characterized by precision, control, and efficiency in motor skills, is essential for safe and productive engagement in physical endeavors.

Developing movement competency involves understanding the principles of movement mechanics and applying these insights to enhance performance. This integrated approach ensures that individuals not only perform movements effectively but also understand the underlying principles that govern efficient movement patterns.

The Science of Human Movement

The human body's ability to move smoothly and effectively is a result of the relationship between specialized systems. The skeletal system provides a sturdy framework, anchoring muscles that create movement. Simultaneously, the muscular system generates

the force necessary for various activities.

These systems not only enable mobility but also contribute to maintaining balance, improving coordination, and ensuring accuracy in daily tasks. Their combined efforts underscore their critical role in supporting our physical capabilities and overall health. Understanding the interaction between these systems helps us recognize the importance of maintaining their health and ensuring they function coordinated. This holistic approach is essential for sustaining optimal movement and functionality throughout our lifespan.

Skeletal System

Could you imagine body without bones?

It would have no shape and would be unable to stand tall. Our vital organs would be completely exposed, making them vulnerable, and we wouldn't be able to move at all. Without our skeletal muscles, our bodies wouldn't have a structure, our delicate organs would be left unprotected, and we wouldn't have the ability to move around

Bone, or osseous tissue, is a hard, dense connective tissue that forms most of the adult skeleton, the support structure of the body. The skeletal system is the body system composed of bones and cartilage and performs the following critical functions for the human body:

1. Protects

Our delicate organs need protection:

- a. The skull protects the brain.
- b. The vertebral column protects the spinal cord.
- c. The rib cage protects the heart and lungs.

2. Supports

Our body needs a framework:

- a. The skeleton gives shape to our bodies.
- b. It holds our vital organs in place.

- c. It enables us to achieve a good posture.

3. Moves

Our muscles use our bones to cause movement:

- a. The skeleton provides attachment for the muscles.
- b. The skeleton is jointed, which allows a wide range of movement.
- c. Different joints allow distinct types of movement.

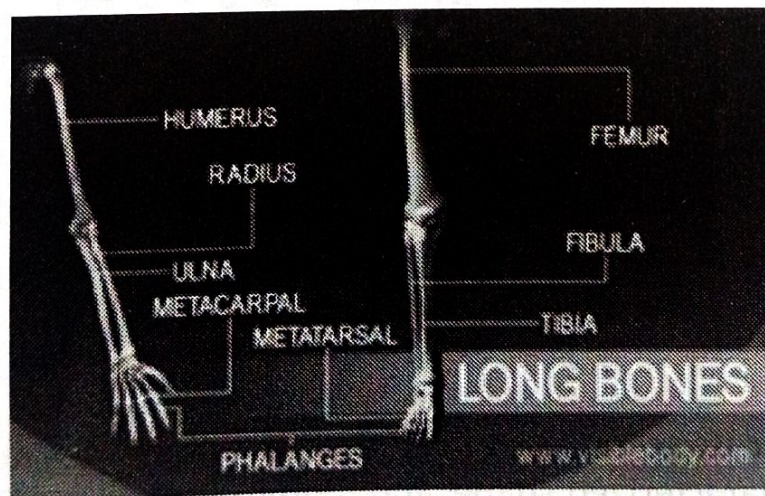
4. Produces blood.

Red and white blood cells are produced in the bone marrow of the ribs, humerus, vertebrae, and femur.

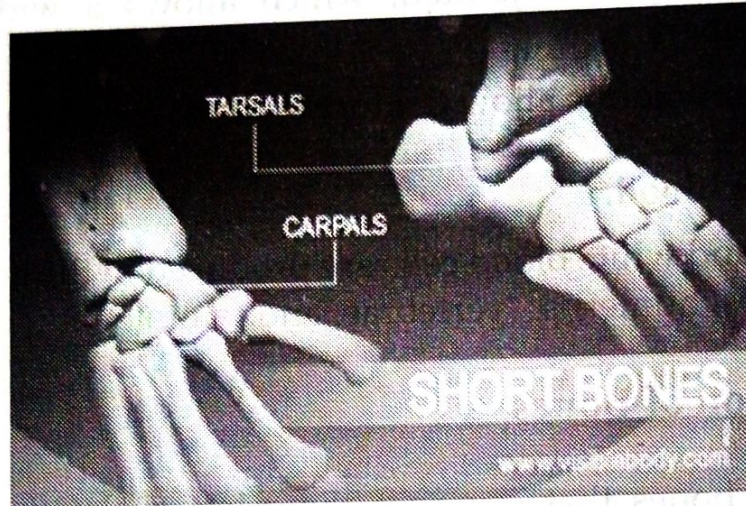
Classification of Bones

The 206 bones that compose the adult skeleton are divided into five categories based on their shapes. Their shapes and their functions are related such that each categorical shape of bone has a distinct function.

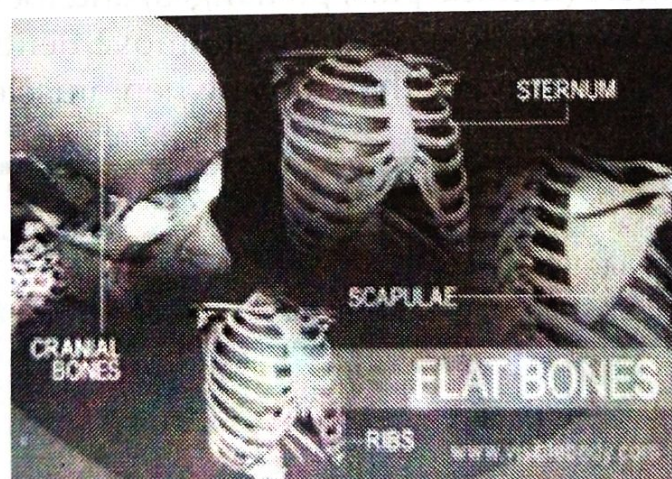
- 1. **Long Bones.** It is a long bone that is cylindrical in shape, being longer than it is wide. Long bones are found in the arms (humerus, ulna, radius) and legs (femur, tibia, fibula), as well as in the fingers (metacarpals, phalanges) and toes (metatarsals, phalanges). We use them in the main movements of our body. Long bones act as levers; they move when muscles contract.



2. **Short Bones.** It is cube-like in shape, being equal in length, width, and thickness. The only short bones in the human skeleton are in the carpals of the wrists and the tarsals of the ankles. We use them in fine movements of our body. Short bones can provide stability and support as well as some limited motion.

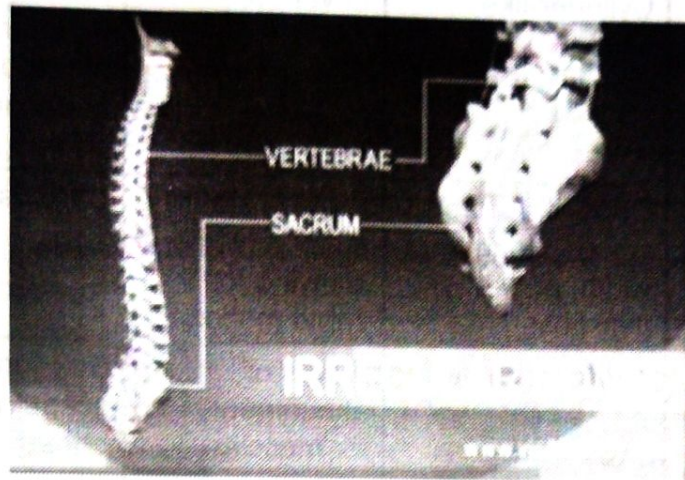


3. **Flat Bones.** A flat bone is typically thin, it is also often curved. Examples include the cranial (skull) bones, the scapulae (shoulder blades), the sternum (breastbone), and the ribs. We use them to protect the organs of our body. Flat bones serve as points of attachment for muscles and often protect internal organs.



4. **Irregular Bones.** It is an irregular bone that does not have any easily characterized shape, and therefore, does not fit any other classification. These bones tend to have more complex shapes,

like the vertebrae that support the spinal cord and protect it from compressive forces. We use them to give our body protection and shape. Many facial bones, particularly the ones containing sinuses, are classified as irregular bones.



5. **Sesamoid Bones.** It is a small, round bone that, as the name suggests, is shaped like a sesame seed. These bones form in tendons (the sheaths of tissue that connect bones to muscles) where a great deal of pressure is generated in a joint. These bones protect tendons by helping them overcome compressive forces. These vary in number and placement from person to person but are typically found in tendons associated with the feet, hands, and knees. The patellae (singular=patella) are the only sesamoid bones found in common with every person. Table 1 reviews bone classifications with their associated features, functions, and examples.



Table 1. BONE CLASSIFICATION

Bone classification	Features	Function(s)	Examples
1. Long	Cylinder-like shape, longer than it is wide	Leverage	Femur, tibia, fibula, metatarsals, humerus, ulna, radius, metacarpals, phalanges
2. Short	Cube-like shape, equal in length, width, and thickness	Provide stability, support, while allowing for some motion	Carpals, tarsals
3. Flat	Thin and curved	Points of attachment for muscles; protectors of internal organs	Sternum, ribs, scapulae, cranial bones
4. Irregular	Complex shape	Protect internal organs	Vertebrae, facial bones
5. Sesamoid	Small and round; embedded in tendons	Protect tendons from compressive forces	Patellae

What are the main parts of our skeleton?

Our skeleton is divided into two parts: **axial skeleton** and the **appendicular skeleton**.

Appendicular Skeleton

1. Shoulder girdle

- a. Made up of two clavicles and two scapulas.
- b. Is only linked by muscles to our vertebral column, this gives us great flexibility in our arms and shoulder. However, it limits the force we can use.

2. Arms

- a. Humerus, radius and ulna
- b. There are eight carpal bones in the wrist.

- c. Five metacarpal bones in the hand
- d. 14 phalanges in each hand

3. Hip girdle

- a. Made up of two halves, each formed by three bones, which are fused together on each side
- b. Forms a very stable joint with the vertebral column and passes the weight of the body to the legs.
- c. Supports the lower abdomen and provides a strong joint for the femur
- d. The female pelvis is wider and shallower than the male pelvis. This is to make childbearing easier, but it does make running less efficient

4. Legs

- a. Femur, tibia and fibula
- b. There are seven tarsals

Axial Skeleton

1. Skull

- a. Made up of 28 bones
- b. There are also 14 bones in the face and six in the ear.
- c. Fused together in early childhood
- d. Protects the brain, eyes and ears
- e. Balance mechanisms found in the ears

2. Sternum

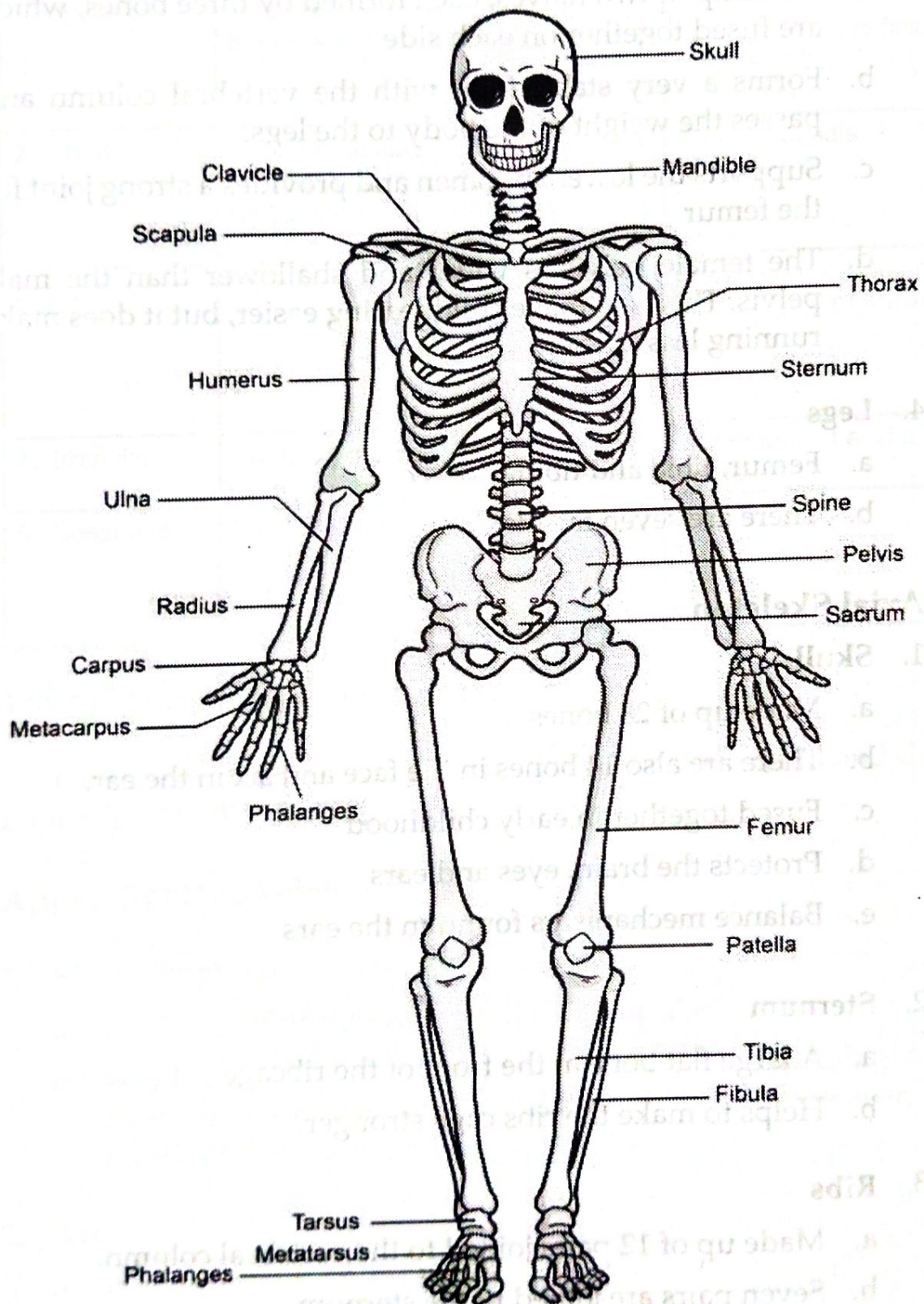
- a. A large flat bone at the front of the ribcage
- b. Helps to make the ribs cage stronger.

3. Ribs

- a. Made up of 12 pairs joined to the vertebral column.
- b. Seven pairs are joined to the sternum.
- c. Three pairs are joined to the seventh rib (false ribs).

- d. Two ribs are unattached (floating ribs).
- e. The ribcage protects our lungs and heart.

Parts of the Skeletal System

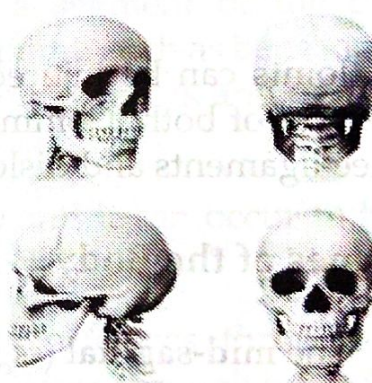


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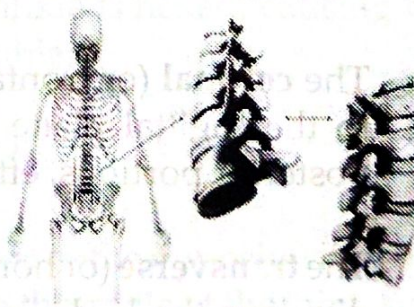
How do we move?

The skeleton has many joints. A **joint**, also called an **articulation**, is any place where adjacent bones or bone and cartilage come together (articulate with each other) to form a connection. There are two classifications of joints: structurally and functionally. Structural classifications of joints consider whether the adjacent bones are strongly anchored to each other by fibrous connective tissue or cartilage, or whether the adjacent bones articulate with each other within a fluid-filled space called a joint cavity. Functional classifications describe the degree of movement available between the bones, ranging from immobile, to slightly mobile, to freely moveable joints. The amount of movement available at a particular joint of the body is related to the functional requirements for that joint. Thus, immobile, or slightly moveable joints serve to protect internal organs, give stability to the body, and allow for limited body movement. In contrast, freely moveable joints allow for much more extensive movements of the body and limbs.

1. **Immovable joints.** A formulation of two bones that have been fused together. Such joints are capable of movement by muscular force. They could as well not exist, except that they do serve a protective function. Slight movement might occur in the joints, resulting in a cushioning effect when external force is applied. *For example*, if a blow is struck on the head, the immovable joints of the cranium will permit slight movement.

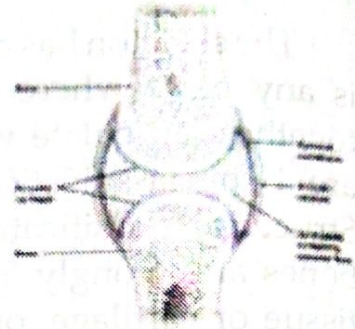


2. **Slightly Movable Joints.** These joints are not fixed as are immovable joints, but the structure of bones and connective tissues in and around the joints restricts the range of motion to only a few degrees. *Examples* of slightly movable joints are those located in the spine. It is also found between the sacrum and ilia, and at



the front and back attachments of the ribs.

3. **Free Movable Joints.** These joints have a comparatively large range of movement and are of prime importance in motor performances. They are in the upper and lower extremities. *Examples* of freely movable joints are the shoulder, elbow, wrist, hip, and knee joints.



Joints and Sports

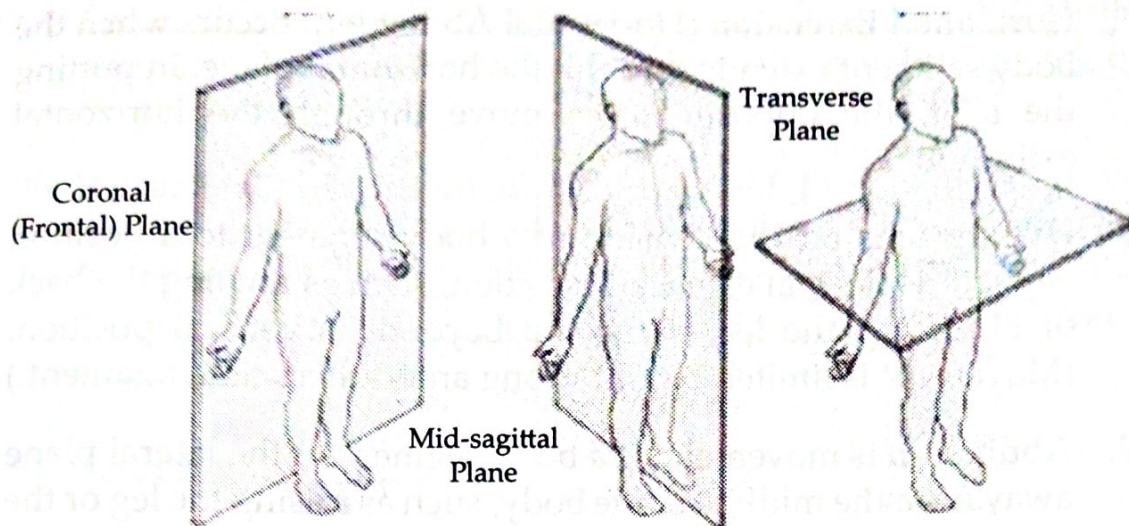
When we make skilled sporting movements, our different joints work smoothly together. To work well, they must be capable of a full range of movement. To give stability to the joint, the muscles and ligaments surrounding each joint must be strong enough.

The demands of sport place severe stress on our joints. We must warm up thoroughly before activity and we should cool down afterwards.

Joints can be injured because of impact, internal forces, or a mixture of both. Common examples include sprained ankle, torn knee ligaments and dislocated shoulder.

Planes of the Body

1. The **mid-sagittal** (*sagitta* is Latin for "arrow") (or median) plane is a vertical plane extending in an anteroposterior direction dividing the body into right and left parts, effectively the forward and backward plane. (A sagittal plane is any plane parallel to the median plane).
2. The **coronal** (or frontal) plane is a vertical plane at right angles to the sagittal plane that divides the body into anterior and posterior portions, effectively the side movement plane.
3. The **transverse** (or horizontal) plane is a horizontal cross-section, dividing the body into upper and lower sections, and lies at right angles to the other two planes: effectively the rotational movement plane.



Kinds of Joint Movements

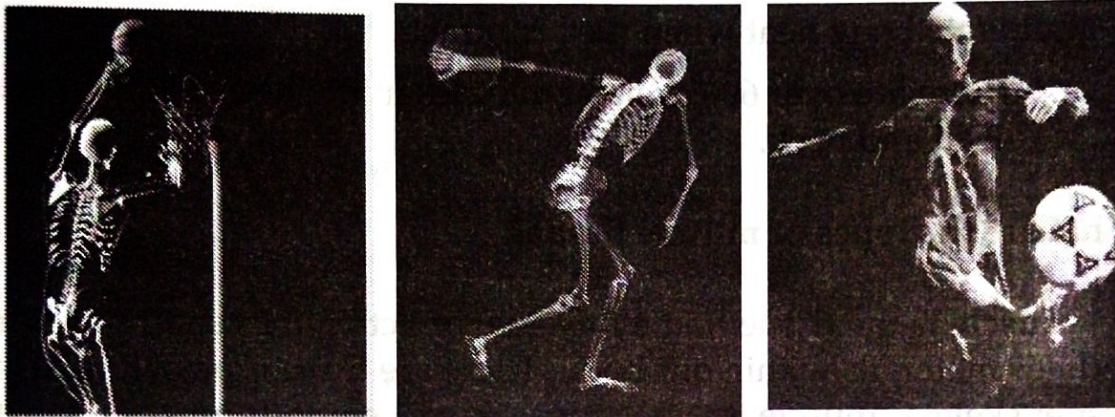
The following are descriptions of all the movements that occur in the body joints. The descriptions assume that the body is in the standard anatomical position, that is, the erect position with the palms forward.

1. **Flexion** (bending) is movement of a segment of the body causing a decrease in the angle at the point, such as bending the arm at the elbow or the leg at the knee. The trunk and neck can flex forward.
2. **Lateral Flexion** is bending sideways, and it can occur to both the right and the left. The trunk and neck can also flex sideways.
3. **Horizontal Flexion** is a body segment flexes through the horizontal plane. *For example*, the arm moves through horizontal flexion at the shoulder joint in throwing the discus or in the sidearm pitch.
4. **Dorsiflex** (Dorsal Flex) is when the ankle is flexed, causing the top of the foot to draw closer to the tibia.
5. **Plantar Flexion** (extension) is the opposite movement at the ankle.
6. **Extension** (Straightening) is movement in the opposite direction of flexion which causes an increase in the angle at the joint, such as straightening the elbow or the knee.

7. **Horizontal Extension** (Horizontal Abduction) occurs when the body segment extends through the horizontal plane. In putting the shot, the opposite arms move through the horizontal extension.
8. **Hyperextension** is extension of a body segment to a position beyond its normal extended position, such as arching the back or extending the leg at the hip beyond its vertical position. (Movement is limited by the strong anterior cruciate ligament.)
9. **Abduction** is movement of a body segment in the lateral plane away from the midline of the body, such as raising the leg or the arm sideways.
10. **Adduction** is movement of a body segment toward the midline, as moving the arm from the outward horizontal position downward to the vertical position.
11. **Rotation** is movement of a segment around its own longitudinal axis. A body segment may be rotated inward (medially) or outward (laterally). The scapula may be rotated upward or downward and the spine may rotate to the right or the left.
12. **Pronation** is rotation of the hand and forearm downward, resulting in a "palm-down" position.
13. **Supination** is rotation of the hand and forearm upward, resulting in a "palm-up" position.
14. **Inversion** is rotating of the foot turning the sole inward.
15. **Eversion** is rotation of the foot turning the sole outward.
16. **Circumduction** is a circular or cone-like movement of a body segment, such as swinging the arm in a circular movement about the shoulder joint. The kind of movement is also possible in the wrist, trunk, neck, hip, shoulder girdle, and ankle joints.
17. **Elevation** is when the shoulder is lifted upward as in shrugging the shoulders.
18. **Depression** is when lowering of the shoulder girdle.

19. **Protraction** (abduction) is movement of the shoulder girdle away from the midline of the body, resulting in broadening of the shoulder.
20. **Retraction** (adduction) is movement of the shoulder girdle toward the midline of the body, resulting in narrowing of the shoulders. The clavicle is capable of some rotation at the sternum and accompanies scapular upward and downward rotation.

Skeletal System and Sports



The whole skeletal system (including bones, joints, ligaments, cartilage, and tendons) contributes to sports performance. Bones protect the body's organs which may have otherwise been damaged, due to contact and force which is involved in most sport. The function of the skeleton in sport is also to provide rigidity and structure to the body, as well as providing strength to the body.

Movement is completely necessary in sport, as well as daily life. The skeletal systems contribution in sport includes the allowance of movement in many different directions, involving distinct groups of bones to do so. Bones also provide leverage which is essential in sport, and they also act as a structure for muscles. Joints aid in movement as well as contributing to sport, as joints allow for movement and the range of flexibility between bones.

Muscular System

All our movements happen because of the shortening (contracting) and lengthening (extending) of muscles. Our muscles can:

1. Enable us to move our body parts.
2. Give us our own individual shape.
3. Protect and keep in place our abdominal organs.
4. Enable us to maintain a good posture.
5. Help in the circulation of our blood.
6. Generate body heat when they contract.
 - a. There are over 600 skeletal muscles in the body – 150 in the head and neck.

What are the types of muscle tissue?

Muscle is the tissue that allows us for active movement of our body or materials within our body. There are three types of muscle tissue: skeletal muscle, cardiac muscle, and smooth muscle. Most of our body's skeletal muscle produce movement by acting on the skeleton.

1. Skeletal muscles

Skeletal or voluntary muscles work as we instruct them. They are under our control. They make our bodies move. We use them for everyday and sporting activities such as walking, running, and jumping.

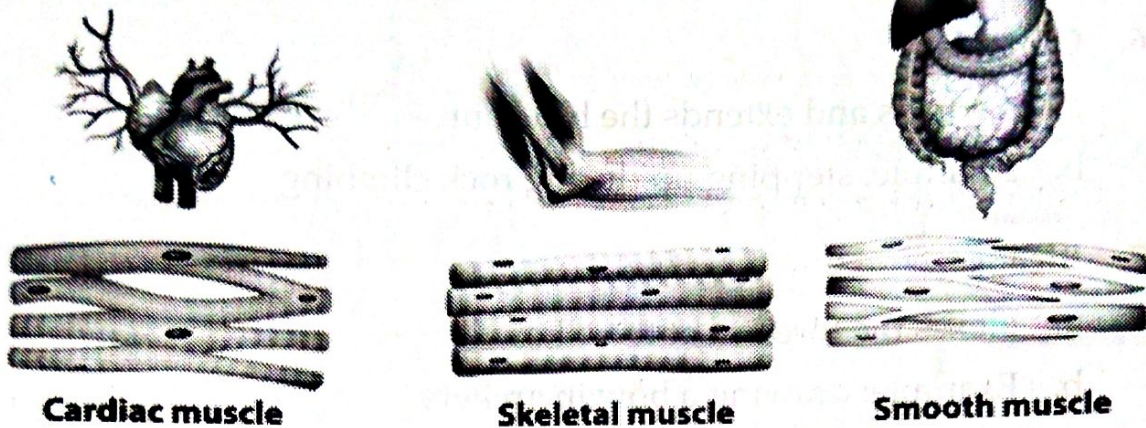
2. Smooth muscles

Smooth or involuntary muscles work automatically. They are not under our conscious control. They work our internal organs such as the stomach, gut, and bladder.

3. Cardiac muscle

Cardiac or heart muscle is a special type of involuntary muscle. It is found only in the heart. It contracts regularly, continuously and without tiring. It works automatically but is under constant nervous and chemical control.

Types of Muscle



What are the main parts of our muscles?

1. Deltoid:

- a. Moves the arm in all directions at the shoulder.
- b. Example: bowling in cricket.

2. Triceps:

- a. Extends the forearm at the elbow.
- b. Extends the arm at the shoulder.
- c. Example: a smash in badminton.

3. Hamstrings:

- a. Extends the hip joint.
- b. Flexes the knee joint.
- c. Example: drawing the leg back before kicking a ball.

4. Trapezius:

- a. Helps to control the shoulder girdle.
- b. Example: a player holding his head up in a rugby scrum.

5. Latissimus dorsi:

- a. Adducts and extends the arm at the shoulder
- b. Example: swimming the butterfly stroke

6. Gluteals:

- a. Abducts and extends the hip joint.
- b. Example: stepping up during rock climbing

7. Biceps:

- a. Flexes the forearm at the elbow
- b. Example: drawing a bow in archery

8. Abdominals:

- a. Rotates and raises the trunk.
- b. Strengthens the abdominal wall.
- c. Helps with breathing.
- d. Example: performing upward circles on the bar in gymnastics

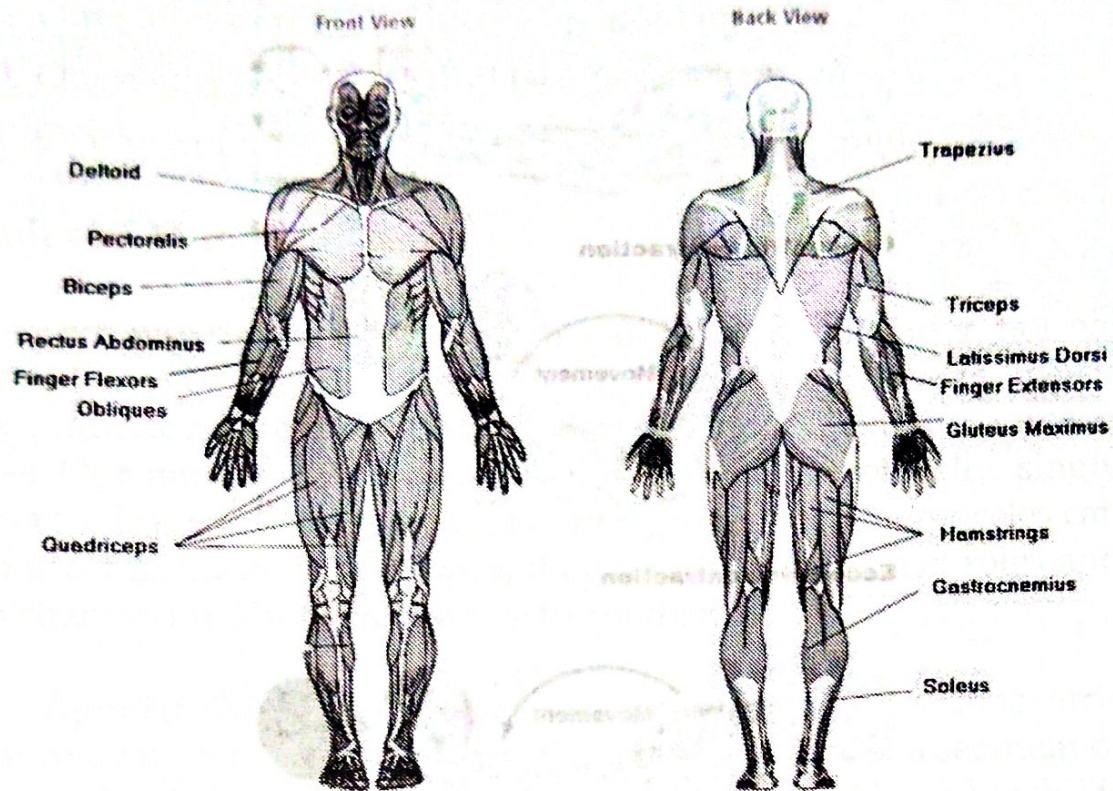
9. Pectorals:

- a. Adduct the arm and shoulder.
- b. Used for deep breathing.
- c. Example: playing a forehand drive in tennis

10. Quadriceps:

- a. Flexes the hip joints.
- b. Extends the knee joint.
- c. Example: taking off in high jump

Major Muscles Group of the Human Body



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How do our muscles work?

Our muscles can work in separate ways. There are three main types of muscular contraction:

1. Isotonic and concentric
2. Isotonic and eccentric
3. Isometric

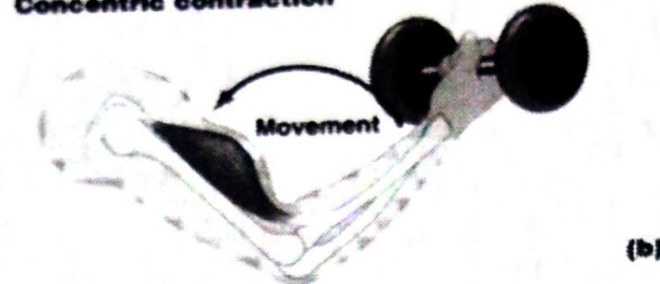
The term '**muscle contraction**' refers to the development of tension within the muscle. There are three main types:

1. In **isometric, or static contraction**, the muscle develops tension with no change in overall muscle length, as when holding a dumbbell stationary in a biceps curl.
2. In **concentric contraction**, the muscle shortens as tension is developed, as when a dumbbell is raised in a biceps curl.
3. In **eccentric contraction**, the muscle develops tension while it lengthens, as in the lowering movement in a biceps curl.

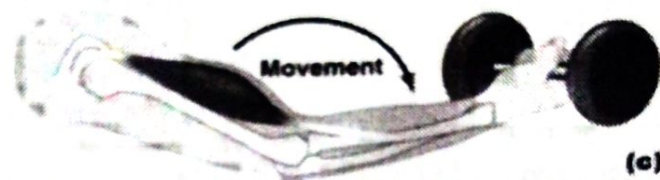
Isometric contraction
Muscle contracts
but does not shorten



Concentric contraction



Eccentric contraction



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Both concentric and eccentric contractions can, theoretically, be at constant tension (isotonic) or constant speed (isokinetic). However, most contractions normally involve neither constant tension nor constant speed.

What happens to our muscular system as we exercise?

1. There is an increased flow of blood to the working muscles.
2. Muscles take up more of the oxygen from the blood.
3. The muscles contract more often and more quickly.
4. More of the muscle fibers contract.
5. There is a rise in temperature in the muscles.
6. Our stores of adenosine triphosphate (ATP) and creatine phosphate (CP) in the muscles are used up.
7. Waste products such as carbon dioxide and lactic acid build up in the muscles.

8. These waste products may lead to tiredness and cramp (muscle fatigue).
9. Our stores of muscle glucose are used up.
10. Our ability to carry on will be affected.
11. Overuse of muscles can lead to soreness and strains.

Roles of Muscles

Our muscle can only contract or relax, and under normal conditions contraction results only from a series of nerve impulses. Our muscle may contract fully or partially, with maximum force or less. Our muscle may contract isometrically or isotonicly, singly (in rare instances) or as a member of a group. Because muscles can contract in these diverse ways, they could act in distinct roles and to change quickly from one role to another.

Agonist (Mover). A muscle is a mover when its concentric contraction contributes to the desired movement of a segment of the body. For instance, in flexion at the elbow, the biceps brachii is a mover.

Antagonist. A muscle is antagonistic to a movement when it must relax to allow the movement to occur. Antagonist muscles cause actions opposite those caused by the agonist muscles. For instance, the triceps brachii is antagonistic to flexion at the elbow.

Agonist and Antagonist Skeletal Muscle Pairs

Agonist	Antagonist	Movement
Biceps brachii: in the anterior compartment of the arm	Triceps brachii: in the posterior compartment of the arm	The biceps brachii flexes the forearm, whereas the triceps brachii extends it.
Hamstrings: group of three muscles in the posterior compartment of the thigh	Quadriceps femoris: group of four muscles in the anterior compartment of the thigh	The hamstrings flex the leg, whereas the quadriceps femoris extend it.

Flexor digitorum superficialis and flexor digitorum profundus: in the anterior compartment of the forearm	Extensor digitorum: in the posterior compartment of the forearm	The flexor digitorum superficialis and flexor digitorum profundus flex the fingers and the hand at the wrist, whereas the extensor digitorum extends the fingers and the hand at the wrist.
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Stabilizer (Fixator). In order for a segment of the body to move, the body part on which the segment moves must possess the right amount of stability. For instance, when the arm moves at the shoulder joint, the shoulder girdle must be held firm by the contraction of certain muscles which attach to it.

Neutralizer. A muscle plays the role of neutralizer when it equalizes or nullifies one or more actions of another muscle. To neutralizes each other, two muscles must cause opposite movements. For instance, the pectoralis major and the latissimus dorsi muscles are both movers in adduction of the humerus; in addition, the pectoralis major flexes the humerus while the latissimus dorsi extends it. When the two muscles neutralize each other's functions of flexion and extension, the result is pure adduction.